# AS-Depth: Unleashing OFA Collaboration to Accelerate Metric Depth Completion on SoC

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***Abstract*—Depth completion reconstructs dense metric depth from sparse or noisy measurements captured by low-cost sensors, offers great benefits to autonomous driving and robotics. However, industrial robotic systems often benefit from real-time metric depth inference, which requires the model to be deployed on a resource-constrained platform (e.g., SoC) with limited computing power and energy budget. Moreover, existing depth completion models are computationally expensive, thus most of the research on depth completion evaluates on data centers GPUs, while the targeted SoC has much smaller computation budget. We observe that the backbone of most depht completion models have clear separation between the encoder and decoder stages, which naturally lends itself to heterogeneous acceleration by offloading one stage to a dedicated accelerator.Building on this insight, we propose as-depth, a heterogeneous system dedicated for accelerating depth completion on SoCs via three techniques: (1) a depth aware QAT strategy and accelerator implemented on FPGA to maintain model accuracy under low precision, (2)an asynchronous pipeline to overlap encoding and decoding stages (3)a heterogeneous execution framework that efficiently partitions the computation between the SoC and the FPGA. Extensive experiments show that as-depth achieves end to end throughput of 30 fps while maintaining the near SOTA metric-depth accuracy. Compared with baselines run on single-GPU, AS-Depth achieves a 5.4× speedup and xxx times improvement in energy efficiency. Our method further demonstrates that a complete model need not reside on one processor: a specific module can be offload to another device, which provides a reusable path for accelerating compatible modules from other models.**

***Index Terms*—architecture & system design, depth completion, heterogeneous computing, SoC, real-time inference**

Application teaser and depth-completion pipeline

Raw RGB-D with missing/noisy depth → completed metric depth → robotic perception

Include a compact Thor–FPGA partition inset

Figure reserved for the advisor-review draft

Fig. 1. Depth completion repairs missing and unreliable measurements from commodity RGB-D sensors before metric depth is consumed by downstream robotic perception. The inset previews the encoder–decoder partition used by AS-DEPTH.

## I. INTRODUCTION

Metric depth measures the distance of each pixel in an image from the camera, providing geometry cues such as spatial and shape information that the 2D color image cannot provide.

TABLE I
DEPTH-COMPLETION FRAME RATES ON THE LISTED PLATFORMS. THE 30 FPS ENTRY FOR AS-DEPTH IS A DESIGN TARGET, NOT A MEASUREMENT.

| | CDM | PriorDA | PromptDA | OMNI-DC | Ours |
|---|---|---|---|---|---|
| Platform | RTX 4090 | RTX 4090 | RTX 4090 | RTX 3090 | SoC |
| FPS | 6.62 | 6.49 | 5.32 | 4.26 | 30 |
| Quality | - | - | - | - | - |

Such depth information can be inputted into the backbone model of applications such as autonomous driving and robotics shown in Figure 1, which greatly improves the capability of spatial reasoning and enables precision interaction with the objects.

With the widespread availability of commodity depth cameras, acquiring depth information appears to be low cost and straightforward. However, commodity RGB-D cameras frequently suffer from missing values, depth holes, flying pixels, and measurement noise, especially under challenging materials such as transparent, reflective, and specular surfaces. Although recent studies have integrated depth into robotic manipulations, performance remain limited by the poor quality of depth data produced by such device. Therefore, evaluations of robotics is typically restricted to simulators where clean depth is available. Thus, depth completion emerges as an effective method for generating dense metric depth from commodity depth cameras, and has become a critical component for bridging the Sim to Real gap.

Recent studies 1234 have demonstrated advanced accuracy of depth completion, but their increasingly large Transformer-based architectures impose substantial computation costs. The selected GPU baselines in Table I range from 4.26 to 6.62 FPS on RTX 3090/4090 platforms. Because the models and platforms differ, these figures provide context rather than a controlled speedup comparison. Achieving real-time depth completion on an edge SoC such as NVIDIA Jetson AGX Thor remains challenging under its computation budget, motivating algorithm and hardware acceleration.

To address this, we propose As-Depth, a asynchronous heterogeneous acceleration system built on SoC and FPGA tailored for depth completion task on edge platforms. As-Depth achieves efficient real time rendering speed and SOTA rendering quality on the edge platform. At the architecure level, it decouples the image representation encoding from the

metric depth decoding stage, enabling these two tasks to be executed asynchronously on the SoC and FPGA according to their computational characteristics. On the algorithmn-hardware co-design level, our depth aware quantization and customized decoder design effectively preserve the accuracy of metric depth while reducing computational overhead. At the accelerator level, our depth aware accelerator integrates an ethernet interface unit and a depth decoding engine to handle the decoding pipeline, enabling efficient data communication and computation.

Building on these insights above, we develop a novel heterogeneous acceleration system, As-Depth, that combines the strengths of both the SoC and the FPGA to achieve real-time depth completion on edge platforms. It leverages the mature CUDA cores and Tensor Cores of the SoC to accelerate the transformer-based encoder, while offloading the CNN decoder to our customized FPGA implementation. This design ensures the maximum utilization of both the SoC and the FPGA, enableing pipelined execution to hide the latency of the offloaded operations. Through our quantization aware training and depth aware customized precision, our system strike a balance between performance and resource overhead. Our contributions are as follows:

- We propose a novel heterogenous system and combine task level asynchronous execution for depth completion based on the existing depth completion pipeline, efficiently mapping computing units according to the different computation characteristics of the SoC and FPGA, and using asynchronous optical queued communication to improve inference performance.
- We design a depth-aware quantization strategy, and a customized precision scheme for the fpga implementation. Our approach successfully balance the accuracy of metric depth between model inference latency.
- We implement a prototype of our As-Depth system on the Nvidia Jetson AGX Thor and Intel Stratrix FPGA platform. Our prototype runs at real-time speed of 30FPS, demonstrating the feasibility of our heterogeneous acceleration approach.
- We implement and evaluate As-Depth on Synopsys. Experiments show that our system achieves 5.4x speedup and xxx improvement in energy efficiency over the latest SOTA implementation on the targeted SoC with the whole depth completion pipeline. To the best of our knowledge, this is the first work that achieves real-time 30FPS inference of depth completion on edge platforms. Our system enables real-time depth near sensor possible which contribute to mitigate the Sim to Real gap in the future rbotics applications. Moreover, our approach demonstrates the potential for heterogeneous acceleration in resource-constrained environments, paving the way for future advancements in edge AI systems.

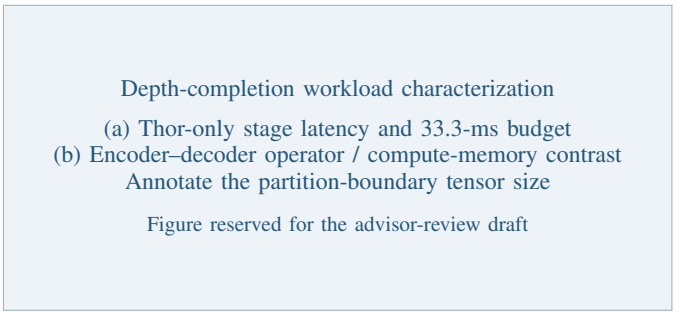



Fig. 2. Workload characterization of the selected depth-completion model. The final plot will report measured Thor-only stage latency and contrast the computational characteristics of the Transformer encoder and CNN decoder under one controlled inference configuration.

## II. Background and Workload Characterization

### A. Depth Completion and the Encoder–Decoder Pipeline

Depth completion estimates a dense metric-depth map from an RGB image and an incomplete depth observation. The input depth may contain holes, outliers, and material-dependent noise, whereas the output must retain both absolute scale and local geometry for downstream robotic perception. Modern methods commonly pair a representation encoder with a dense prediction decoder. The encoder extracts global visual and geometric features; the decoder fuses and upsamples these features to the target depth resolution.

AS-Depth preserves this algorithmic decomposition and uses its boundary as a system partition. The selected depth-completion network and its trained weights are the workload being accelerated, while this paper focuses on the low-precision deployment, heterogeneous execution, and dedicated decoder hardware. The final manuscript will state the exact backbone, decoder, input/output resolutions, and feature tensor exported at the partition.

### B. Computational Characteristics

The two stages expose different opportunities for edge acceleration. The encoder contains Transformer blocks whose matrix operations and MLP layers can use Thor's mature GPU and Tensor Core execution stack. The decoder is composed primarily of structured dense-prediction operators for which customized parallelism, buffering, and precision can be implemented on FPGA. This difference motivates a stage-level mapping rather than repeatedly moving individual operators between devices.

The argument must be grounded in measurements. We therefore profile the unmodified model on Thor and report the latency of preprocessing, encoder, decoder, and postprocessing under a fixed resolution, precision, batch size, and power mode. We additionally characterize the dominant operator classes, their arithmetic intensity or compute/memory demand, and the size of the encoder–decoder boundary tensor.

### C. Edge SoC–FPGA Heterogeneous Execution

An edge SoC offers a complete software environment and high-throughput tensor units, but its compute and energy

budget is shared with sensing, planning, and control. An FPGA provides configurable precision, dataflow, and memory organization for a stable subgraph. The combination is useful only when the partition is coarse enough to limit communication and when execution across devices is scheduled to hide, rather than merely add, transfer latency.

Prior GPU depth-completion systems establish the accuracy of large models, and prior FPGA accelerators demonstrate the efficiency of specialized low-precision datapaths. AS-DEPTH connects these directions by retaining the encoder on the SoC, specializing the decoder on FPGA, and treating the model boundary and cross-frame schedule as first-class parts of the design.

## III. MOTIVATION

Based on the workload characterization above, we identify two challenges in deploying real-time depth completion on a heterogeneous edge platform. Each challenge exposes an opportunity that directly motivates one part of AS-DEPTH's design.

*a) Challenge 1: Monolithic execution misses the real-time budget.:* The Transformer encoder and CNN decoder have different operator mixes and hardware preferences, yet a conventional deployment executes both stages sequentially on one processor. The resulting end-to-end latency includes the full encoder, decoder, and runtime path for every frame. Merely moving the decoder to another device does not remove this serialization if communication and accelerator execution remain on the critical path.

*b) Opportunity 1: Task-level heterogeneous pipelining.:* This opportunity follows from the clear stage separation. The clear encoder–decoder boundary permits a coarse-grained partition: the Thor executes the encoder while the FPGA executes the decoder. Across a stream of frames, queued communication and double buffering can overlap the encoder of frame $i+1$ with transfer and decoding of frame $i$. This turns device specialization into steady-state pipeline throughput rather than an additional synchronous offload cost.

*c) Challenge 2: End-to-end low precision can degrade metric depth.:* Low precision is beneficial on both sides of the partition. In particular, selected MLP computation in the Thor-side encoder and the FPGA-side decoder can be executed at reduced precision. A uniform quantization policy, however, must simultaneously accommodate Transformer activation distributions, depth-regression sensitivity, and the numerical contract at the cross-device boundary. Aggressive quantization can therefore save computation and bandwidth while introducing scale errors or local depth artifacts.

*d) Opportunity 2: Coordinated depth- and hardware-aware precision.:* This opportunity addresses the tension between efficiency and depth quality. Rather than treating encoder quantization, feature transfer, and decoder quantization as independent optimizations, AS-DEPTH coordinates their precision choices. Depth-aware quantization (DAQ) targets the layers and signals that most affect metric-depth accuracy, while hardware-aware precision choices map the encoder MLP and decoder datapath to the efficient modes of Thor and FPGA, respectively. The same boundary representation also reduces the cost of cross-device feature transfer.

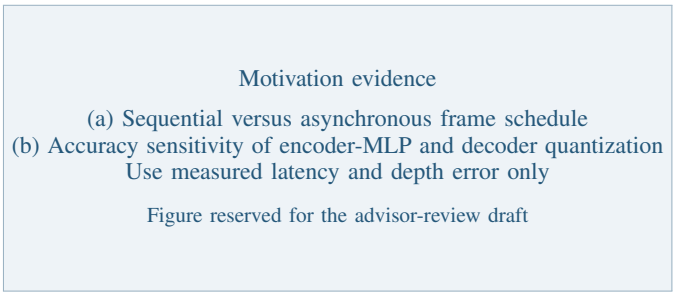



Fig. 3. Evidence motivating AS-DEPTH. (a) A synchronous offload leaves communication and decoder execution on the critical path, whereas a task-level schedule overlaps work across frames. (b) Low-precision choices in both the encoder MLP and decoder must be coordinated to avoid excessive metric-depth error.

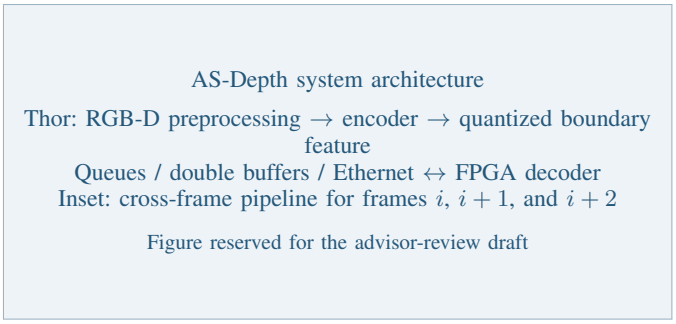



Fig. 4. Overview of AS-DEPTH. Thor executes the representation encoder and produces the cross-device feature tensor; the FPGA executes the depth decoder. Queues and double buffers preserve frame order while overlapping encoding, communication, and decoding across consecutive frames.

## IV. AS-DEPTH DESIGN

AS-DEPTH combines three mechanisms derived from the opportunities in Section III. A task-level execution framework partitions the model and overlaps consecutive frames; DAQ coordinates low precision across the Thor encoder and FPGA decoder; and a dedicated decoder accelerator turns the regular decoder workload into an efficient hardware dataflow.

### A. Task-Level Asynchronous Heterogeneous Execution Framework

*a) Model partition and tensor contract.:* The model is partitioned once at the encoder–decoder boundary. Thor consumes the RGB image and sensor depth, executes the encoder, and exports a quantized feature tensor. The FPGA consumes this tensor and returns the completed metric-depth map. The final paper will state the exact tensor shape, layout, precision, scale metadata, and frame identifier used by this interface.

*b) Queued cross-frame execution.:* The host runtime assigns separate buffers to encoder output, outbound transfer, decoder input, and completed depth. Producer–consumer queues make ownership explicit and prevent a buffer from being reused before its consumer finishes. After pipeline fill,

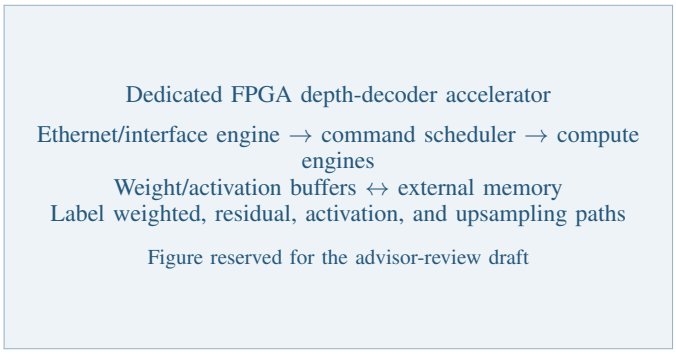



Fig. 5. Architecture of the dedicated FPGA depth-decoder accelerator. The final figure will show the communication front end, programmable layer controller, weighted and auxiliary compute engines, local reuse buffers, and external-memory datapath.

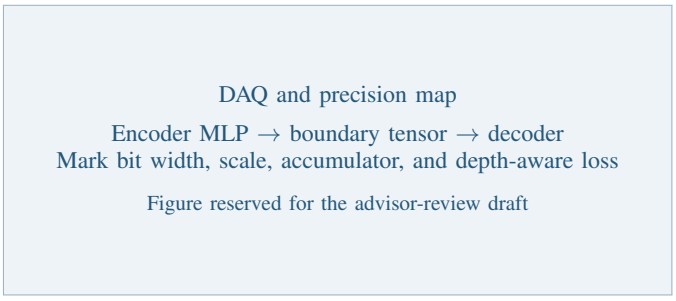



Fig. 6. Depth-aware quantization across the heterogeneous inference path. The final diagram will identify the quantized encoder MLP operations, the cross-device representation, decoder precision, and the depth-domain training signal.

the encoder for the next frame can progress in parallel with transfer and decoder execution for the current frame. We report both single-frame latency and steady-state throughput because only the latter benefits fully from cross-frame overlap.

*c) Communication and back pressure.:* The runtime batches feature metadata with the boundary tensor and preserves frame order on the return path. If either device becomes the bottleneck, bounded queues apply back pressure instead of silently dropping or reordering frames. Section V measures transfer time separately from useful computation so that communication hiding can be verified.

### B. Depth-Aware Quantization and Customized Precision

DAQ treats quantization as an end-to-end property of the heterogeneous model. On Thor, it targets selected MLP computation in the encoder; at the partition boundary, it defines the representation transmitted to the FPGA; and inside the FPGA, it determines the precision of decoder weights, activations, accumulators, and output depth. Training optimizes the quantized path against metric-depth supervision so that numerical error is weighted by its effect on the final depth map rather than by feature reconstruction alone, extending the standard quantization-aware training setting [1].

*Draft note: Specify the encoder MLP layers covered by DAQ, the exact bit widths, per-tensor or per-channel scaling, the depth-aware loss, calibration procedure, and any layers retained at higher precision. The "without DAQ" ablation must keep the same partition and target precision while replacing DAQ with a conventional quantization policy.*

### C. Dedicated FPGA Depth Decoder Accelerator

The FPGA accelerator specializes the weighted and auxiliary operators in the depth decoder while retaining a programmable layer schedule. Its datapath must expose the precision contract established by DAQ, reuse weights and activations on chip where profitable, and stream larger intermediate tensors through external memory without changing model semantics. The communication front end converts received feature packets into accelerator-visible buffers and returns the completed depth map after the final layer. This design follows the broader principle of mapping quantized networks to customized FPGA dataflows [2] while specializing the execution path for metric-depth decoding.

*a) Compute and dataflow.:* The architecture description will quantify parallel multiply–accumulate lanes, accumulator precision, supported convolution variants, and the mapping of activation, residual, and upsampling operators. A layer controller selects the appropriate engine and coordinates local buffers with external memory.

*b) Control and correctness boundary.:* For every layer, the scheduler records tensor dimensions, addresses, quantization parameters, and dependencies. Verification compares each hardware layer boundary and the final depth map against the quantized software reference before end-to-end performance is reported.

## V. Evaluation

We evaluate AS-DEPTH by answering four questions. First, can heterogeneous execution sustain the 30-FPS real-time target once communication and runtime overheads are included? Second, does DAQ preserve metric-depth quality on challenging sensor data? Third, how does the complete system compare with recent depth-completion methods? Finally, what FPGA resource, communication, power, and energy costs are required to obtain these gains?

### A. Methodology

*a) Baselines.:* We use two groups of baselines for different purposes. The controlled system comparison runs the same depth-completion network in three deployments: *GPU-only* on an RTX 4090, *Thor-only* on the target edge SoC, and AS-DEPTH with its encoder on Thor and decoder on FPGA. The GPU-only and Thor-only baselines execute the complete model sequentially on one device; AS-DEPTH includes feature transfer and output reconstruction in its end-to-end time. For the component study, AS-DEPTH without DAQ retains the same partition, asynchronous runtime, accelerator, and target precisions, but replaces the proposed depth-aware policy with conventional low-precision quantization. This makes the comparisons in Table II attributable to deployment and quantization choices rather than to different networks.

For model-level context, we additionally compare against CDM, PriorDA, PromptDA, and OMNI-DC, following the methods introduced in Section I. We use their released checkpoints and official evaluation procedures whenever available. Because their published FPS values come from different GPU platforms, the state-of-the-art throughput plot labels the execution platform of every method and does not treat cross-platform FPS as a controlled speedup claim.

*b) Benchmarks.:* We evaluate all accuracy-oriented experiments on HAMMER and ClearPose. These datasets exercise different object geometries and difficult sensing conditions, so results are reported for each dataset rather than averaged into one score. Every method uses the same test split, validity mask, crop, input resolution, and depth range within a dataset. When a released method requires a different input representation, only the documented preprocessing is changed; the ground-truth mask and metric computation remain common. The system and ablation experiments use batch size one, which matches continuous RGB-D capture on the target robotic platform.

*c) Experiment platform.:* The prototype comprises an NVIDIA Jetson AGX Thor and an Intel Stratix FPGA connected through the system interface described in Section IV-A. The FPGA clock, external-memory configuration, link rate, Thor power mode, and software versions are held fixed across the two AS-DEPTH variants. The RTX 4090 baseline uses the same model weights, input resolution, preprocessing, and output validation. Before submission, the final platform table will state the exact CUDA/TensorRT stack, FPGA device and achieved clock, host–FPGA link, and precision assigned to every model stage.

*d) Metrics and measurement protocol.:* We measure both single-frame latency and steady-state throughput. Latency is decomposed into preprocessing, encoder execution, boundary-tensor transfer, FPGA decoding, return transfer, and output reconstruction; FPS is measured only after the asynchronous pipeline is filled. Timing includes synchronization needed for a valid completed-depth output, rather than reporting kernel time alone. Each result will state the warm-up length, measured frame count, and run-to-run variation.

Depth quality is evaluated with AbsRel, RMSE, MAE, and $\delta_{1.05}$. Lower is better for the three error metrics and higher is better for $\delta_{1.05}$. We report system power, energy per completed frame, and FPS/W under a consistent measurement boundary. FPGA resource results use post-fit utilization and achieved frequency rather than pre-synthesis estimates.

### B. Ablation Experiments

Figure 7 evaluates the four configurations in Table II. We interpret the bars cumulatively: the single-device baselines establish the deployment cost, the system without DAQ exposes the benefit of heterogeneous execution, and the complete system shows whether depth-aware precision recovers quality without giving up the hardware path.

*a) Throughput and heterogeneous execution.:* Figure 7(a) reports absolute FPS and marks the 30-FPS real-time threshold. Thor-only versus AS-DEPTH without DAQ is the primary controlled comparison for the Thor–FPGA partition: both process the same frames with the same network, while only the latter offloads the decoder and overlaps encoder, communication, and decoder work across frames. We therefore report not only their FPS difference, but also the stage-latency breakdown and the fraction of transfer hidden by the asynchronous schedule. AS-DEPTH with and without DAQ is then compared under the same runtime to determine whether the precision policy changes throughput. GPU-only is retained as a high-end performance reference, not as an edge-device energy baseline. The completed discussion will state absolute FPS first, followed by speedup over Thor-only, and will explain any remaining gap using the slowest pipeline stage rather than attributing it generically to "offloading."

TABLE II
CONFIGURATIONS USED FOR THE CONTROLLED SYSTEM COMPARISON AND COMPONENT ABLATION. "CONVENTIONAL LP" AND "DAQ" USE THE SAME TARGET BIT WIDTHS; THEY DIFFER ONLY IN THE QUANTIZATION POLICY.

| Configuration | Encoder | Decoder | Precision policy |
|---|---|---|---|
| RTX 4090 only | RTX 4090 | RTX 4090 | reference |
| Thor only | Thor | Thor | reference |
| AS-DEPTH w/o DAQ | Thor | FPGA | conventional LP |
| AS-DEPTH w/ DAQ | Thor | FPGA | depth-aware LP |

*b) Metric-depth accuracy.:* Figures 7(b)–(e) separate HAMMER and ClearPose and show both an error metric and a threshold accuracy metric. The decisive comparison is AS-DEPTH without versus with DAQ because their model partition, target bit widths, accelerator, and runtime are identical. Any quality difference therefore measures the value of coordinating the quantized encoder MLP, boundary representation, and FPGA decoder using metric-depth supervision. The GPU-only and Thor-only results provide the unpartitioned reference and reveal the total degradation introduced by the heterogeneous low-precision path. We report each dataset independently and discuss whether improvements come from reducing large depth errors, increasing the fraction of accurate pixels, or both; a single aggregate "near-SOTA" statement is not used in place of these measurements.

### C. Comparison with State-of-the-Art Methods

*a) Throughput.:* Figure 8(a) compares the reported inference rate of CDM, PriorDA, PromptDA, OMNI-DC, and AS-DEPTH. Each bar is annotated with its execution platform and the plot includes the 30-FPS reference line. This comparison answers a deployment question–whether current depth-completion methods can service a real-time sensor stream–while the controlled GPU-only/Thor-only comparison above supports architectural speedup claims. In the completed result discussion, we report AS-DEPTH's measured end-to-end FPS and identify whether its advantage comes from the coarse model partition, cross-frame overlap, or low-precision execu-

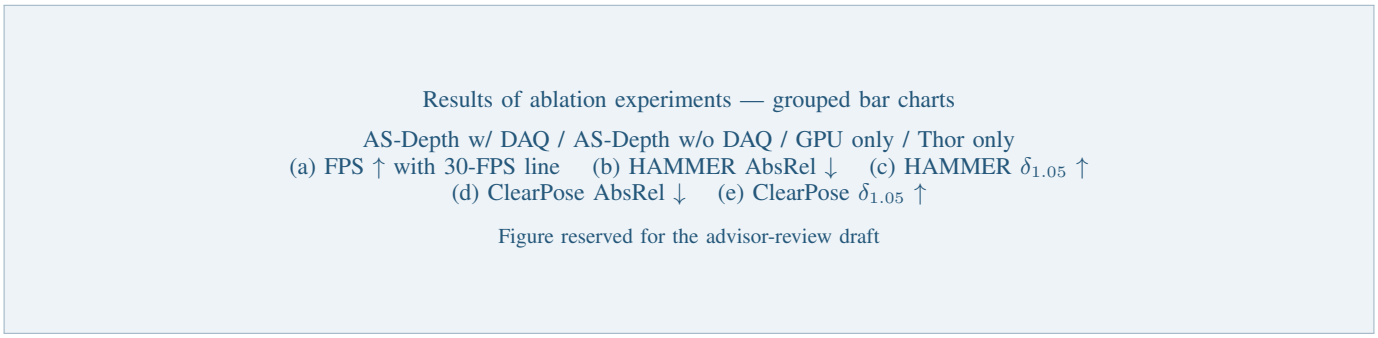



Fig. 7. Results of the component ablation. Throughput isolates the effect of heterogeneous execution, while the HAMMER and ClearPose panels quantify the accuracy recovered by DAQ under the same partition and target precision.

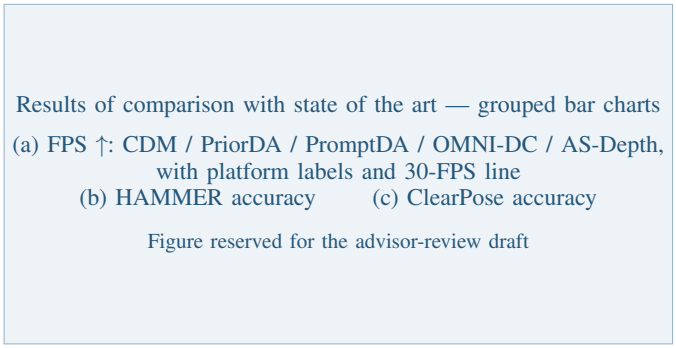



Fig. 8. Comparison with state-of-the-art depth-completion methods. Throughput is labeled by platform and provides deployment context; accuracy is evaluated on HAMMER and ClearPose with a common protocol.

tion, using the ablation evidence rather than the cross-platform bars alone.

*b) Depth accuracy.:* Figures 8(b) and (c) compare all methods under common HAMMER and ClearPose protocols. The discussion reports the absolute metric for every method, AS-DEPTH's gap to the best-performing method, and its rank on each dataset. We then relate that gap to the deployment result: the relevant question is whether AS-DEPTH reaches real-time edge throughput while keeping the accuracy loss small and consistent across both datasets. Claims of state-of-the-art or near-state-of-the-art quality will be made only for metrics and datasets for which the measured values support them.

### D. Overhead Analysis

*a) FPGA resource and communication overhead.:* Table III reports post-fit logic, register, DSP, and on-chip-memory utilization together with achieved frequency. To connect resource cost to the system design, we also report boundary bytes per frame, measured link latency, and standalone decoder latency. These measurements show whether the selected precision fits the target device and whether feature transfer is sufficiently small to be hidden behind Thor encoding. Resource percentages are always reported with the device capacity and synthesis tool settings so that raw counts are interpretable.

TABLE III
POST-FIT HARDWARE, COMMUNICATION, AND ENERGY OVERHEAD. VALUES MARKED "–" AWAIT IMPLEMENTATION OR MEASUREMENT.

| Metric | Value | Metric | Value |
|---|---|---|---|
| Logic / registers | – | DSP blocks | – |
| On-chip memory | – | Frequency | – |
| Feature bytes/frame | – | Link latency | – |
| Decoder latency | – | System power | – |
| Energy/frame | – | FPS/W | – |

*b) Power and energy efficiency.:* Power is measured for Thor-only, the active FPGA, and the complete Thor–FPGA system using the same input stream and steady-state window used for FPS. We derive energy per frame as system power divided by completed frames per second and additionally report FPS/W. The analysis separates the extra power introduced by the FPGA and link from the energy saved by shortening the end-to-end execution time. RTX 4090 power is reported as a separate reference because its platform boundary differs from the embedded prototype. The final discussion will pair every energy-efficiency ratio with both its power and throughput components, explaining when additional hardware power is offset by a shorter end-to-end execution time.

## VI. CONCLUSION

This paper presents AS-DEPTH, a heterogeneous depth-completion system that decouples representation encoding on Thor from depth decoding on FPGA. Its task-level runtime overlaps consecutive frames, while DAQ coordinates low-precision encoder MLP computation, the cross-device representation, and the dedicated decoder datapath. The final manuscript will close by reporting the measured throughput, metric-depth accuracy, and energy efficiency achieved by this combination on HAMMER and ClearPose.

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